

From: Peter Gent
Sent: Fri, 18 Feb 2022 13:13:52 -0700
To: Peter Gent
Subject: Cassia Crossbills in Colorado

----- Forwarded message -----

From: **Craig Warren Benkman** <CBenkman@uwyo.edu>
Date: Thu, Nov 18, 2021 at 9:58 AM
Subject: Re: Cassia Crossbills in Colorado
To: Christian Nunes <christianalfrednunes@gmail.com>

Hi Christian,

Thank you for writing. Someone from Idaho just sent me your announcement from the Colorado Birds listserv. That is amazing but unfortunately as you note it is not too surprising given the fire in 2020 that burned ~25% of the pine forest for Cassia Crossbills. Moreover, and possibly more relevant, based on the number of days with temperatures reaching or exceeding 90 F in 2020 (none in 2021), I anticipate that the Cassia population will decline further by about 20% in each of the following four years starting last winter. Not good news for them.

Fires obviously occurred in the past (probably each stand burned on average every 125 years or so based on our estimates), but they were apparently much smaller in part because the matrix of sagebrush in which stands of lodgepole pine are embedded in the South Hills was not invaded by cheatgrass. The 2020 fire spread quickly through sagebrush (and cheatgrass), whereas that was unlikely to occur decades ago so that fires would burn a smaller proportion of the forest patches in any given year. And now the fires are occurring more frequently - nearly all the recent fires since I started working there in 1996 have been caused by or are suspected to have been caused by hunters not putting out their campfires or by ATVs.

I am not sure what you mean by their population structure. The population has undoubtedly declined in proportion to the forest lost and I suspect is declining further. They declined by about 80% between 2003 and 2011 because of a series of hot summer days (more than occurred in 2020), but other than undergoing a decline their population didn't seem to change. What I suspect based on our previous studies is that the rare cases of hybridization in the South Hills will become even rarer, however, the opportunity for hybridization outside the South Hills/Albions will increase.

It is hard for me to say whether those birds you found bred in or outside the South Hills/Albions. Middle of July is a period when roaming Cassia Crossbill families are quite common. If you had found

them in say March or September, then I would be fairly confident they didn't nest in the South Hills/Albions. Thanks again for sharing.

Best,

Craig

From: Christian Nunes <christianalfrednunes@gmail.com>

Date: Thursday, November 18, 2021 at 9:28 AM

To: Craig Warren Benkman <CBenkman@uwyo.edu>

Subject: Cassia Crossbills in Colorado

□ This message was sent from a non-UWYO address. Please exercise caution when clicking links or opening attachments from external sources.

Greetings Dr. Benkman,

I thought you would be interested in my report of Cassia Crossbill from Grand County, Colorado, on July 16, 2021. Tim Spahr and Matt Young just confirmed my initial ID. Recordings of various calls and some songs can be viewed here: <https://ebird.org/checklist/S91855481>

These tracks sat on my recorder until last week. I had a casual conversation with another crossbill aficionado and remembered that I had recorded these birds over the summer. I downloaded them off my memory card thinking they were Type 5s, but lo and behold!

This adds an interesting footnote to the Cassia Crossbill story. It's maybe not terribly surprising that some have wandered far and wide after the fire in the South Hills. It's probably not the first time fire has altered the ecosystem, either. It's interesting to think about what the birds must do during these

post-fire periods. Will their population structure be maintained? The birds in my recording were acting like a family unit, with an adult male, adult female, and a streaky juvenile. Did they breed outside Cassia County? I could speculate all day...

All the best,

Christian Nunes

Longmont, Colorado